Business Model Canvas

Timar | A Digital Platform for Mangrove Restoration on the Coast of Karachi

Note: during the Final Year Project, Timar is an academic, non commercial system. The revenue streams below are routes available to Saltwater Marina afterwards, not income for the student project. The platform produces carbon evidence for a verifier, it does not issue carbon credits.

DESIGNED FOR	DESIGNED BY	SUPERVISOR	VERSION
Timar / Saltwater Marina	**Aftab Ahmed Samoo (2312398) · Haseeb Mudasir (2313403)**	**Sir Abid Ali**	**1.0**

KEY PARTNERS

Saltwater Marina: the partner organisation. Provides field access, plantation sites, volunteers and the photographs the models learn from.

Copernicus, European Union: free Sentinel-2 satellite imagery of the Karachi coast, refreshed every few days.

Global Mangrove Watch: reference maps of existing mangrove extent and change.

Sindh Forest Department: receives forwarded cutting complaints and is a potential data partner.

WWF Pakistan and IUCN: domain guidance on species, sites and restoration practice.

Cloud and GPU providers: free Colab and Kaggle tiers for training, free tier hosting for the platform.

Schools, universities and corporate teams: a steady source of volunteers, trip bookings and sponsorship.

KEY ACTIVITIES

Building and running the platform: plot registry, growth monitoring, trips, sponsorship, volunteers, complaints and the admin portal.

Collecting and labelling field photographs with Saltwater Marina, a dataset that does not exist anywhere today.

Training and evaluating five models: plant health, seedling counting, site suitability, satellite change detection and survival risk.

Monthly satellite processing: pulling imagery, computing the vegetation index, comparing months and raising alerts.

Verifying alerts and survey records on the ground so every published number can be defended.

KEY RESOURCES

Two full stack developers sharing every part of the system.

The survival record the platform itself creates, plot by plot, month by month.

Saltwater Marina's field teams, plantation sites and local knowledge.

Free Sentinel-2 imagery and open mangrove extent maps.

Trained models, the spatial database and the growing photograph archive.

VALUE PROPOSITIONS

For Saltwater Marina: one system in place of WhatsApp groups and personal phones. Every plot mapped, every count recorded, every report one click away.

For sponsors and companies: proof instead of promises. Coordinates, a live survival counter and dated photographs for the plots they funded.

For volunteers: a clear way in. Apply, receive real field work, and keep a record of what you contributed.

For citizens: finally understand what mangroves do for Karachi, visit them, sponsor a seedling and report cutting when you see it.

For the coast: cutting becomes visible within weeks instead of years, and new planting goes where survival is most likely.

Honest by design: the system states plainly that it produces carbon evidence, not carbon credits, and that a photograph assesses health rather than guaranteeing survival.

CUSTOMER RELATIONSHIP

Self service: anyone can open the public dashboard, read the awareness content and book a trip without being contacted.

Assisted and personal: field coordinators approve volunteers, assign tasks and answer media and interview requests.

Long term tracking: a sponsor's page keeps updating for months after they pay, which is what turns one donation into the next.

Community: a feed where people share field photographs and discuss the coast.

Transparency as the relationship: headline numbers are public with no login, so trust does not depend on asking.

CHANNEL

The web platform itself, usable on a phone and installable like an app.

Saltwater Marina's existing social media presence and volunteer network.

Plantation drives and eco tourism trips, where visitors become contributors.

School, university and corporate programmes.

Word of mouth from sponsors sharing their own plot page.

CUSTOMER SEGMENTS

Community restoration organisations: Saltwater Marina first, then any group doing the same work on another coastline.

Volunteers: students and residents who want to do something physical about the climate rather than only read about it.

Individual sponsors: people buying seedlings for themselves, as gifts, or in someone's memory.

Companies: corporate responsibility and climate teams that need local impact they can actually evidence.

Eco tourists and schools: groups who want to visit the creeks and see the forest.

Government and researchers: the Sindh Forest Department and universities studying the Indus Delta.

Donors and grant bodies funding coastal restoration and blue carbon work.

COST STRUCTURE

Development: a two person effort across eight months, covering thirteen modules and five models.

Training compute: free Colab and Kaggle GPU tiers, under twenty GPU hours in total.

Satellite data: free from Copernicus. The real cost is processing time, not the imagery.

Hosting and database: free tier during the project, a modest paid tier once real users arrive.

Object storage for field photographs, which grows with every survey round and is the largest long term cost.

Data labelling: team time spent annotating photographs for the health and counting models.

Field costs: travel to the creeks, absorbed into Saltwater Marina's existing plantation drives.

REVENUE STREAM

Seedling sponsorship: individuals and families pay per seedling and follow their plot. The main public facing stream.

Corporate sponsorship packages: a company funds a plot and receives verified, branded impact reporting.

Eco tourism trips: ticketed creek visits, which Saltwater Marina already has the capability to run.

Platform licensing: other restoration organisations pay to run their own instance with their own sites and species.

Grants and climate funding: verified survival data makes the organisation eligible for funding that is currently out of reach.

Carbon project support: fees for preparing the evidence pack a verification body requires. Timar prepares evidence, it does not issue credits.